

Abstract

The Key–Lock Model is a discrete-time, spatially local dynamical system on a finite grid. Its state consists of non-negative amounts of finitely many *types*, some of which are grouped into rigid, mobile *bundles*. Every type is specified by the same data: two vectors in $\mathbb{R}^d$ (a *key* and a *lock*), a *product* type, and a *valence*. All interactions between types are mediated by a single bilinear quantity, the affinity $A(q, r) = k_q \cdot l_r$, through four mechanisms: binding-limited diffusion of free amounts, mass-action reactions, threshold-based bundle cohesion, and valence-weighted motion of bundles. No interaction is specified pairwise; consequently the model is invariant under any relabeling of types that preserves the affinity structure, and under any change of basis of $\mathbb{R}^d$ that preserves the key–lock pairing.

This document states the axioms, defines the model's state space and transition map, proves that the transition map satisfies the axioms, describes how runs of the model proceed, and lists the questions the model leaves open.

The model in brief

The world is a grid of cells. Each cell holds amounts of various *types*: kinds of thing, such as what one might call a signal, a material, a body part, or a feeling. The model draws no such distinctions; all types are described the same way.

Every type carries two arrows, a **key** and a **lock**. How well one type's key points along another type's lock, their **affinity**, is the only way types ever influence each other. A strong positive affinity is a **fit**; a strong negative one is a **clash**. Fits and clashes drive everything:

- **Loose material spreads** from cell to cell, but is held back where the contents of a cell fit it.
- **Material reacts**: when a key fits a lock, one type is converted into another; when a key clashes with a lock, both are consumed.
- **Bundles**, groups of material that move as one unit, hold together exactly as long as their contents fit each other.
- **Bundles move** toward or away from cells containing what their contents respond to.

The goal of the model is that complex behavior (sources, trails, seeking, erosion, breakage) arises from these few uniform rules, rather than from rules written for particular kinds of thing.

How to read this document

Every axiom, definition, and result is stated formally, then followed by:

- **Meaning**: what it says in plain terms, and why it is there;
- **Example**: where useful, a small concrete case.

Proofs follow the results they prove and end with $\square$.

0. Conventions

Notation	Meaning
$\mathbb{R}_{\geq 0}$	the non-negative reals
$[P]$	Iverson bracket: 1 if the statement P is true, else 0
$ S $	cardinality of a finite set S
$u \cdot w$	standard inner product on $\mathbb{R}^d$: $u \cdot w = \sum_m u[m] w[m]$
$x + \delta$	componentwise addition in $\mathbb{Z}^2$
$\ y - x\ _1$	taxicab distance: $ y_1 - x_1 + y_2 - x_2 $
$\emptyset, \perp$	the empty set; a symbol meaning "none" or "unowned"

All sums are finite. A sum over an empty index set is 0, and a minimum over an empty set is 1.

Example. $[2 > 1] = 1$ and $[2 < 1] = 0$. For $u = (1, 2)$ and $w = (3, -1)$, $u \cdot w = 1 \cdot 3 + 2 \cdot (-1) = 1$.

Part I. Axioms

The axioms come in two groups. **Structural axioms** (S1–S6) fix the objects the model is built from: the grid, time, the types, what each type is, what a state of the world is, and the constants. **Dynamical axioms** (A1–A10) constrain how a state may change from one tick to the next. Between the two groups, Section I.2 defines the derived quantities needed to state the dynamical axioms.

Meaning. The structural axioms say *what exists*. The dynamical axioms say *what any law of change must respect*. Parts III–VI then give the specific laws, and prove that they respect the dynamical axioms.

I.1 Structural axioms

Axiom S1 (Space). The world is a finite grid

$$\mathcal{W} = \{1, \dots, m\} \times \{1, \dots, n\} \cong \mathbb{Z}^2.$$

Elements of $\mathcal{W}$ are *cells*. The *neighborhood* of a cell x is

$$\mathcal{N}(x) = \{y \in \mathcal{W} \mid \|y - x\|_1 = 1\}.$$

The relation $y \sim(x)$ is symmetric and irreflexive, and $|\mathcal{N}(x)| \leq 4$.

Meaning. Space is a rectangle of m columns and n rows of cells. Each cell touches the cells directly north, south, east, and west of it, and no others. "Symmetric" means that if y touches x , then x touches y ; "irreflexive" means no cell is its own neighbor.

Example. With $m = 3$, $n = 2$, the cell $(1, 1)$ is a corner with $\mathcal{N}((1, 1)) = \{(2, 1), (1, 2)\}$; the cell $(2, 1)$ lies on an edge with $\mathcal{N}((2, 1)) = \{(1, 1), (3, 1), (2, 2)\}$. In a larger grid, an interior cell has four neighbors.

Axiom S2 (Time). Time is discrete, $\mathbb{N}$. The state at $t + 1$ is obtained from the state at t by one application of the transition map (Part II), possibly preceded by external input (Axiom A10).

Meaning. The world advances in steps called *ticks*. Nothing happens between ticks except optional intervention by the player.

Axiom S3 (Types). There is a finite, non-empty set T of *types*, containing a distinguished element a , the *ambient*.

Meaning. A type is a kind of thing that can be present in the world, in some amount. Every kind of thing is a type; there are no separate categories for signals, materials, or creatures. The ambient is a special type that fills the whole world as a fixed background.

Example. $s = \langle s, w, h \rangle$, where one might informally think of s as a scent, w as wall material, and h as a hunger. The model itself attaches no such meanings.

Axiom S4 (Type data). There is a fixed integer $d \geq 1$, and each type q is assigned a tuple

$$q = (k_q, l_q, p_q, v_q) \in \mathbb{R}^d \times \mathbb{R}^d \times (T \cup \perp) \times \{-1, 0, +1\}$$

called its *key*, *lock*, *product*, and *valence*. The ambient has $p_a = \perp$. No type has the ambient as its product. The tuples do not change over time.

Meaning. Everything about how a type behaves is contained in four entries:

Entry	What it is	Role
key k_q	an arrow in $\mathbb{R}^d$	how q acts on other types
lock l_q	an arrow in $\mathbb{R}^d$	how other types act on q
product p_q	another type, or none	what q turns into, or produces, when changed by a reaction
valence v_q	+1, 0, or −1	whether q , inside a moving bundle, pulls it toward (+1) or away from (−1) what q 's lock fits

The key and lock are independent: a type may act on others in one way and respond to something entirely different. The space $\mathbb{R}^d$ in which they live is called *signature space*; its coordinates have no meaning in themselves (Proposition VIII.2). Products are never the ambient, so that no reaction can create more ambient (Axiom A3).

Example. With $d = 2$, the tuple $s = (0, 1), (1, 0), \perp, +1$ describes a type whose key points along the second axis, whose lock points mostly along the first, which produces nothing when it reacts, and which pulls toward what its lock fits.

Axiom S5 (State). Fix a countably infinite set X of *bundle labels*. A *state* is a triple $\omega = (x, y, z)$ with

$$x, y \in \mathbb{R}_{\geq 0}, \quad z \in T \cup \perp$$

written $q(x), p(x), (x)$, subject to:

1. $\langle x \rangle = 1$ for all x ,
2. $\langle x \rangle = 0$ for all x ,
3. $\langle x \rangle = 0$ whenever $\langle x \rangle = \perp$.

$q(x)$ is the *free amount* of q at x . $\langle x \rangle$ is its *bundled amount*. $\langle x \rangle$ is the bundle that *owns* x , or $\perp$. The set of all states is Σ .

Meaning. A state is a complete snapshot of the world. In each cell, every type may be present in two forms:

- **free:** loose material, which spreads on its own;
- **bundled:** material belonging to the bundle that owns the cell, which moves only with that bundle.

$\langle \cdot \rangle$ records which bundle, if any, owns each cell. The three conditions say: the ambient is present at amount 1 everywhere and only in free form; and a cell with no owner holds no bundled material.

Example. Let $\Sigma = \{x_1, x_2\}$ and $\Sigma = \{q, w\}$ with

	q	w	$\langle \cdot \rangle$	$\langle \cdot \rangle$	$\langle \cdot \rangle$
x_1	1	0.5	0	0	3
x_2	1	1.0	0	0	0

Cell x_1 is owned by bundle γ which holds 3 units of w there; both cells contain free q ; x_2 is unowned, so it holds no bundled material.

Axiom S6 (Constants). There are fixed constants

$$\alpha > 0, \quad 0 < \beta \leq 1, \quad \gamma > 0, \quad \delta > 0, \quad \epsilon > 0,$$

called the *dead zone*, *hop rate*, *binding sensitivity*, *reaction rate*, and *bond threshold*.

Meaning. Five numbers, the same for every type and every cell, set the scale of every law:

Constant	Controls	Used in
α	how strong an affinity must be to count at all	Definition 3
β	how fast free material spreads when nothing holds it	Part III
γ	how strongly being held slows spreading	Part III
δ	how fast reactions proceed	Part IV
ϵ	how strongly a bundle must hold together to survive	Part V

None is determined by the model; they are chosen, like the type data.

Example. $\alpha = 0.1$, $\beta = 0.5$, $\gamma = 1$, $\delta = 0.1$, $\epsilon = 0.5$ the values used in Part X.

Remark 1.1 (Bundles). Each label $\langle x \rangle \in \Sigma$ is a *bundle*, with cell set $\delta^{-1}(\langle x \rangle)$. Since δ is a function, distinct bundles have disjoint cell sets, and each cell holds the bundled amounts of at most one bundle. Writing

$\tilde{g}(x) = g(x)$ for $x \in \mathcal{X}$ recovers the usual description of a bundle as a set of cells carrying amounts. The representation by a single function g is equivalent and is used throughout.

Meaning. A bundle is simply the set of cells sharing one owner label, together with the bundled material in them. Because each cell has exactly one owner or none, two bundles can never overlap, and so one function g suffices to record all bundled material.

Example. If $\mathcal{R} = (\mathcal{V}, \perp, \cdot)$ on four cells $x_1, \dots, x_4$ then $\mathcal{S} \models x_1, x_2$ and $\mathcal{S} \models x_4$ and x_3 belongs to no bundle.

I.2 Derived quantities

All quantities below are functions of a state $\mathcal{S} = (\mathcal{V}, \mathcal{R})$ and the fixed type data.

Definition 1 (Total amount). $g(x) = q(x) + g(x)$.

Meaning. The total amount of q in a cell, counting both free and bundled forms.

Example. In the example of S5, $g(x_1) = 0 + 3 = 3$ and $g(x_1) = 0.5$

Definition 2 (Affinity). For $q, r \in \mathcal{Q}$

$$A(q, r) = k_q \cdot l_r.$$

In general $A(q, r) \neq A(r, q)$. The quantity $A(q, q) = k_q \cdot l_q$ is the *self-affinity* of q ; it is not constrained by any axiom.

Meaning. Affinity measures how well q 's key lines up with r 's lock. Geometrically, $k_q \cdot l_r = |k_q| |l_r| \cos \theta$ where θ is the angle between them: it is large and positive when the arrows point the same way, zero when they are perpendicular, and negative when they point in opposite directions. It is directional: $A(q, r)$ compares q 's key with r 's lock, while $A(r, q)$ compares r 's key with q 's lock. Self-affinity compares a type's key with its own lock, and since these are independent entries of its tuple, a type need not fit itself.

Example. Let $k_q = (1, 0)$, $l_q = (0, 0.5)$, $k_r = (0, 2)$, $l_r = (2, 1)$. Then

$$A(q, r) = 1 \cdot 2 + 0 \cdot 1 = 2, \quad A(r, q) = 0 \cdot 0 + 2 \cdot 0.5 = 1, \quad A(q, q) = 0.$$

q affects r more strongly than r affects q , and q does not fit itself at all.

Definition 3 (Fit and clash).

$$A^+(q, r) = \max(0, A(q, r)), \quad A^-(q, r) = \max(0, -A(q, r)).$$

For every pair, at most one of $A^+(q, r)$, $A^-(q, r)$ is positive.

Meaning. Small affinities are ignored, so that types do not all weakly affect one another. The *fit* A^+ is how far affinity rises above 0; the *clash* A^- is how far it falls below 0. Between 0 and 0 (the *dead zone*), both are zero and the pair is indifferent. Subtracting rather than simply discarding small values, makes fit and clash grow continuously from zero, so a small change in a vector never causes a sudden jump in behavior. Every law of the model uses fit or clash, never raw affinity.

Example. With $\mathcal{S} = 0.2$:

$A(q, r)$	$A^+(q, r)$	$A^-(q, r)$	Relationship
15	13	0	strong fit
05	03	0	weak fit
015	0	0	indifferent
-06	0	04	moderate clash

Definition 4 (Binding).

$$q(x) = \sum_r \nu_r(x) A^+(q, r).$$

Meaning. Binding measures how strongly the contents of cell x hold on to one unit of q . Each type r present contributes its amount times how well q 's key fits r 's lock. It counts everything in the cell, free and bundled, including the ambient and q itself. Binding slows the spreading of free q (Part III).

Example. Suppose x holds $q(x) = 2$ and $w(x) = 3$, with $A^+(q, q) = 02$, $A^+(q, w) = 05$ and $A^+(q, \emptyset) = 0$. Then

$$q(x) = \underbrace{2}_{qB} \underbrace{02}_{BB} + \underbrace{3}_{wB} \underbrace{05}_{BB} + \underbrace{1}_{\emptyset B} \underbrace{0}_{BB} = 19$$

Definition 5 (Response).

$$q(x) = \sum_r \nu_r(x) A^+(r, q).$$

For a bundle ν , the *response excluding* is

$$\check{q}(x) = \sum_r \nu_r(x) - [q(x) = \sum_r \nu_r(x) A^+(r, q)].$$

Meaning. Response measures how much of what q 's lock fits is present in cell x : every type whose key fits q 's lock contributes, weighted by its amount. It runs in the opposite direction to binding:

	Uses	Measures
Binding $q(x)$	q 's key against others' locks	how strongly the cell holds q
Response $q(x)$	others' keys against q 's lock	how much in the cell q responds to

Response also measures how fast q would absorb things in that cell (Proposition IV.9). The excluded version omits a bundle's own material, so that a bundle deciding where to move is not attracted to itself (Remark VI.5).

Example. Let a cell contain $q(x) = 2$, with $A^+(q, q) = 09$ and $A^+(q, w) = 05$ and let it be owned by a bundle ν holding $q(x) = 1$ with $A^+(w, q) = 05$ Then

$$q(x) = 2 \cdot 09 + 1 \cdot 05 = 28 \quad \check{q}(x) = 2 \cdot 09 + 1 \cdot 05 = 23.$$

Definition 6 (Cohesion, link, connection, bond strength). For a bundle ν , a cell x and a neighbor y $q(x)$ is

$$\begin{aligned} \ell(x) &= \sum_{y \in \mathcal{N}(x)} \ell(x, y) A^+(y, x), & \ell(x, y) &= \sum_{z \in \mathcal{N}(x)} \ell(x, z) \ell(y, z) A^+(z, x), \\ \ell(x, y) &= \ell(x, y) + \ell(y, x), & \sigma &= \sum_{x \in \mathcal{V}} \ell(x) + \sum_{x \in \mathcal{V}} \sum_{y \in \mathcal{N}(x)} \ell(x, y). \end{aligned}$$

These are the *cohesion*, *link*, *connection*, and *bond strength*. All are non-negative, and ℓ is symmetric.

Meaning. These measure how strongly a bundle's material holds together:

- **Cohesion** $\ell(x)$: how strongly the bundle's material within one cell holds itself together. Multiplying two amounts counts the unit-to-unit pairings between them, as in counting handshakes; each pairing contributes its fit.
- **Link** $\ell(x, y)$: how well the material in x fits the material in a neighboring cell y , in that direction.
- **Connection** $\ell(x, y)$: both directions together, the total strength of attachment between two neighboring cells.
- **Bond strength** σ : the whole bundle's internal fit, adding every cell's cohesion and every link.

Only touching material is counted: cells that are not neighbors are connected only through the cells between them.

Example. Let $\mathcal{B}$ own two neighboring cells x, y and hold a single type w with $A^+(w, w) = 0.8$ with $\mathbf{x}(x) = 2$ and $\mathbf{x}(y) = 1$:

Quantity	Calculation	Value
$\ell(x)$	$2 \cdot 2 \cdot 0.8$	3.2
$\ell(y)$	$1 \cdot 1 \cdot 0.8$	0.8
$\ell(x, y)$	$2 \cdot 1 \cdot 0.8$	1.6
$\ell(y, x)$	$1 \cdot 2 \cdot 0.8$	1.6
$\ell(x, y)$	$1.6 + 1.6$	3.2
σ	$3.2 + 0.8 + 1.6 + 1.6$	7.2

Definition 7 (Local bond). For a bundle $\mathcal{B}$ and $x \in \mathcal{V}$ and $v \in \mathcal{V}$:

$$\ell(x) = \sum_{y \in \mathcal{N}(x)} \ell(x, y) A^+(y, x) + A^+(x, x).$$

Meaning. The local bond asks: does anything in the bundle, in this cell or a neighboring cell of the same bundle, hold type v , or is held by it? It adds up fit in both directions between x and the bundle's nearby material. If it is zero, nothing in the bundle holds v at that spot, and x will be released (Definition V.1).

Example. A bundle cell holds $\mathbf{x}(x) = 1$ with $A^+(w, w) = 0.8$ and $\mathbf{x}(x) = 0.1$ of a type u whose fits with every type are zero. With no neighboring bundle cells,

$$\ell(x) = 1 \cdot (0.8 + 0.8) = 1.6 \quad \ell(x) = 0.$$

w is held; u is not.

Definition 8 (Drive). A type q is a *drive* of bundle i if $q \neq \emptyset$ and $q(x) > 0$ for some $x \in \mathbb{Z}^2$.

Meaning. A drive is a type with a pull, toward or away, present somewhere in a bundle. Only bundles with drives move (Part VI).

Example. A bundle holding $y > 0$ of the type j in the example of S4 has j as a drive, since $y = +1$. A bundle containing only types with valence 0 has no drives.

Definition 9 (Displacements). The set of displacements is

$$\Delta = \{(0, 0), (1, 0), (-1, 0), (0, 1), (0, -1)\}.$$

For a bundle i and $\delta \in \Delta$, $\delta = x + \delta \cdot x$. The displacement is *valid* for i if $\delta \in \Delta$.

Meaning. A displacement is one of five possible moves: stay, east, west, north, or south, by one cell. $\delta \in \Delta$ is where the bundle would be after the move: every one of its cells shifts by the same step, so a bundle moves rigidly. A move is valid if no cell would leave the grid.

Example. If $i = \{(1, 1), (2, 1)\}$ and $\delta = (1, 0)$ (east), then $\delta \cdot i = \{(2, 1), (3, 1)\}$. The cell $(2, 1)$ appears in both: the bundle's rear cell moves into where its front cell was. The move is valid if $m \geq 3$.

Definition 10 (Blocked move, facing pairs). A valid $\delta \in \Delta$ is *blocked* for i if

$$F(i, \delta) = \{y \in i \mid y + \delta \in i, y \neq \emptyset\}.$$

Its set of *facing pairs* is

$$F(i, \delta) = \{(x, x + \delta) \mid x \in i, (x + \delta) \in F(i, \delta)\}.$$

Each facing pair consists of neighboring cells owned by different bundles.

Meaning. A move is blocked if any cell the bundle would move into is owned by a different bundle. Since bundles cannot overlap, the move does not happen. The facing pairs are the places where the two bundles press against each other: each is a cell of the moving bundle together with the cell of the other bundle directly in front of it. Nothing ever overlaps; facing cells are side by side.

Example. A bundle i occupies $(1, 1)$ and $(2, 1)$ and tries to move east, while bundle j occupies $(3, 1)$:

row 1:	.	b	b	b'	.
	0	1	2	3	4

Then $\delta \cdot i = \{(2, 1), (3, 1)\}$, $F(i, \delta) = \emptyset$ and $F(j, \delta) = \{(2, 1), (3, 1)\}$. The rear cell $(1, 1)$ is in no facing pair, since its step lands on i 's own cell.

I.3 Dynamical axioms

Let τ denote one tick of the dynamics (excluding external input). The following must hold for every state i and every outcome $\omega \in \Omega$.

Axiom A1 (Locality). Every change of type involves amounts in a single cell or in two neighboring cells. Free amounts move only between neighboring cells. A bundle's cell set changes by translation by an element of $\mathcal{C}$ or by the removal of cells, partition into several bundles, or release of all its cells.

Meaning. Nothing acts at a distance. Reactions happen between material in the same cell or in touching cells; free material moves at most one cell per tick; and bundles move at most one cell per tick, or break apart.

Example. Free material in $(1, 1)$ cannot reach $(3, 1)$ in one tick; it must pass through $(2, 1)$ first.

Axiom A2 (Uniformity). $\mathcal{L}$ depends on the type data only through the values $A(q, r)$, η_q , and ν_q , and treats no type specially except $\perp$. Precisely: $\mathcal{L}$ commutes with every relabeling of types that preserves these values and fixes $\perp$ (Theorem VIII.1).

Meaning. There are no special-case rules. The laws never ask *which* type something is, only how it fits and clashes with others, what it produces, and its valence. The ambient is the single exception.

Example. If two types q and q' have identical fits and clashes with every type (in both directions), identical products, and identical valences, then swapping all amounts of q with all amounts of q' before a tick gives the same result as swapping them after.

Axiom A3 (Ambient). $\eta_{\perp} = 1$ and $\nu_{\perp} = 0$.

Meaning. The ambient is never used up, never created, never moves, and is never bundled. It is an unchanging background with which everything can react.

Axiom A4 (Conservation of flow). The movement of free amounts between cells preserves $\sum_x q(x)$ for every q .

Meaning. Spreading only relocates free material; it never creates or destroys any.

Example. If two cells hold 10 and 0 units of free q and spreading moves 25% from the first to the second, the totals before and after are both 10.

Axiom A5 (Spreading). In every tick, each free amount $q(x)$ with $q \neq \perp$ sends toward each neighbor of x a fraction $(\eta_q(x)) \cdot q(x)$ of itself, where η_q is positive and strictly decreasing.

Meaning. Free material always spreads, by sending a share of itself to each neighbor every tick. The more strongly it is held where it is, the smaller that share. Because η_q is positive, being held slows spreading but never stops it entirely.

Example. With the law of Part III and $\eta_{\perp} = 0$, material held by nothing sends 10 to each neighbor; material with $\eta_q(x) = 2$ sends $10 \cdot 2^{-2} = 2.5$.

Axiom A6 (Reactions). Amounts change type only through reactions between pairs of amounts that share a cell or face each other across a blocked move. The quantity of a reaction depends only on the two amounts involved and their fit or clash, except that it is reduced when an amount would otherwise become negative.

Meaning. The only way one type turns into another is a reaction between two specific amounts, either side by side in one cell, or pressed together across a blocked move. How much reacts depends only on those two

amounts and how well they fit or clash, except when several reactions together would use up more of an amount than exists, in which case they are scaled down.

Axiom A7 (Sources). The total non-ambient amount

$$|x| = \sum_{q \in \mathcal{Q}} \mathbb{P}_q(x)$$

can increase only through absorption in which the first amount of the pair is ambient.

Meaning. New material can enter the world only by drawing on the background. Every other process either relocates material, converts it one-for-one, or destroys it.

Example. A type whose lock fits the ambient's key can produce a steady stream of its product (Example VII.15). Two non-ambient types reacting can never increase the total.

Axiom A8 (Bundles). Bundled amounts change cell only by translation of their bundle. Bundles are never created by $\downarrow$. Apart from translation, a bundle changes only by releasing some of its amounts as free amounts, by partitioning into several bundles, or by releasing all of its amounts. Each such change depends only on fits among the bundle's own amounts and on $\downarrow$.

Meaning. Bundled material stays with its bundle. The laws can break bundles (shed material, split them, dissolve them) but never build new ones; bundles enter the world only through the initial state or the player. Whether a bundle breaks is decided entirely by how well its own material holds together, compared with the threshold $\downarrow$.

Axiom A9 (Motion). Whether a bundle translates, and by which displacement, depends only on the amounts and valences of its drives and their responses to the cells of $\downarrow + \delta$, up to uniformly random tie-breaking and a uniformly random order of processing bundles. A blocked move leaves the bundle in place.

Meaning. A bundle moves according to what its drives want: toward cells containing what attracts it, away from cells containing what repels it, more strongly the more of each drive it holds. Randomness enters only to break ties and to decide which bundle moves first. A bundle can never move into another bundle.

Axiom A10 (External input). Between ticks, the state may be modified by an external agent, only by changing free amounts of non-ambient types, and by adding or removing bundles (setting $\downarrow$ and $\downarrow_n$ on cells, subject to S5).

Meaning. The player can place or remove loose material and bundles between ticks, but cannot touch the ambient or alter the laws.

Part II. The transition map

II.1 Structure

One tick is the composition

$$\downarrow \circ \text{tick} \circ \downarrow$$

where:

Operator	Name	Acts on	Deterministic
$\mathbb{F}$	flow	$\mathcal{J}$	yes
$\mathbb{R}$	reactions	$\mathcal{J}, \mathcal{M}$	yes
$\mathbb{B}$	bundle structure	$\mathcal{J}, \mathcal{M}, \mathcal{I}$	yes, up to the choice of fresh labels
$\mathbb{M}$	motion	$\mathcal{J}, \mathcal{M}, \mathcal{I}$	no

$\mathbb{M}$ uses a uniformly random ordering and uniformly random tie-breaking, so $\mathbb{M}$ is a Markov kernel on $\mathcal{S}$. Given a fixed source of randomness (a seed), $\mathbb{M}$ is a deterministic map.

$\mathbb{F}$ and $\mathbb{R}$ are *simultaneous*: every quantity on the right-hand side of their defining equations is evaluated at the input state.

Meaning. A tick consists of four stages, applied in order (the composition is read right to left):

1. **Flow:** free material spreads (Part III).
2. **Reactions:** material changes type (Part IV).
3. **Bundle structure:** bundles shed material, split, or dissolve if they no longer hold together (Part V).
4. **Motion:** bundles move, or push against one another (Part VI).

The order is deliberate: material spreads before it reacts, so that it can arrive where it reacts; bundles are checked after reactions, because reactions are what weaken them; and bundles move last, as they stand after everything else.

"Simultaneous" means that in flow and reactions, every cell is computed from the same snapshot, so no cell's result depends on the order in which cells are processed. Motion is the one stage that randomness enters, which makes a tick a random process. Fixing the random seed makes it fully reproducible.

II.2 Amounts as coordinates

To state reactions uniformly, index all amounts of a state by

$$\mathbb{A} = \{ (q, x) \mid q \in \mathcal{Q}, x \in \mathcal{X} \} \quad \mathbb{A}^+ = \{ (q, x) \mid q \in \mathcal{Q}^+, x \in \mathcal{X} \}.$$

An element $\zeta = (q, x) \in \mathbb{A}$ is an *amount*; q is its *type* and x its *cell*. We write ζ for (q, x) when ζ is clear. The ambient amounts are $(\emptyset, x)$.

Meaning. This is bookkeeping: each individual amount in the world is named by three things, whether it is free ($\emptyset$) or bundled (ζ), its type, and its cell. It lets a reaction say precisely *which* amount it uses and *which* it adds to. Nothing new is introduced; $\mathbb{A}$ simply lists the entries of $\mathbb{A}$ and $\mathbb{A}^+$.

Example. In the example of S5, the positive amounts are $(\emptyset, x_1)$, $(\emptyset, x_2)$, (q, x_1) , (q, x_2) , and (w, x_1) , with $(w, x_1) = 3$.

Part III. Flow

Definition III.1 (Flow). For $q \in \mathbb{R}^n$ and $x \in \mathbb{R}^n$ let

$$q(x) = q(x) \cap q(x),$$

with q evaluated at the input state. Then q sets

$$\dot{q}(x) = q(x) + \sum_{y(x)} q(y) - q(x),$$

and leaves q and q unchanged.

Equivalently: x sends $q(x)$ to each neighbor and retains the $q(x)$ shares directed outside x .

Meaning. Picture each cell holding a cup of each free type. Every tick, every cup simultaneously pours off four equal portions, one toward each direction, hands each portion to the neighbor in that direction (keeping it if the direction leads off the grid), and receives the portions its neighbors pour toward it. The new amount is what it had, minus what it handed out, plus what it received.

Each portion $q(x)$ is the product of three factors:

- $q(x)$: how much is there to pour;
- q : the largest possible portion, so that at most the fraction q of unheld material leaves per tick, split over four directions;
- $q(x)$: how free q is to leave, between 0 and 1. It equals 1 when nothing holds q , and shrinks toward 0 as binding grows.

$q(x)$	0	05	1	2	∞
$q(x)$	100	06	037	014	0007

Example. Two neighboring cells, $q = 1$, nothing holding q , and $q = (10, 0)$. Each portion is a quarter of the cell's amount. The first cell hands 25 to the second, keeping its other three portions; the second has nothing to give. After one tick, $q = (525, 0)$. (Continued in Example VII.11.)

Proposition III.2 (Non-negativity). $\dot{q}(x) \geq 0$.

Proof. $q(y) \geq 0$, so $\dot{q}(x) \geq q(x) - |x| q(x) \geq q(x) (1 - q(x)) \geq 0$, using $|x| q(x) \leq q(x)$ and $q(x) \leq 1$. $\square$

Meaning. A cell never hands out more than it has, because its four portions together are at most the fraction q of its amount. This is why q may not exceed 1.

Proposition III.3 (Conservation; Axiom A4). $\sum_x \dot{q}(x) = \sum_x q(x)$.

Proof. $\sum_x \sum_{y(x)} q(y) - q(x)$ is a sum over ordered pairs of neighbors. By symmetry of the neighbor relation, each unordered pair x, y contributes $q(y) - q(x) + q(x) - q(y) = 0$. $\square$

Meaning. Every portion that leaves one cell arrives in another, so the total never changes.

Proposition III.4 (Spreading; Axiom A5). The fraction of $q(x)$ sent to each neighbor is $(q(x)) = \frac{q(x)}{d(x)}$, which is positive and strictly decreasing in $q(x)$. $\square$

Meaning. The flow law is one concrete law satisfying Axiom A5.

Proposition III.5 (Uniqueness of the holding factor). Let $f: [0, \infty) \rightarrow (0, 1]$ be continuous with $f(0) = 1$, strictly decreasing, and multiplicative:

$$f(x+y) = f(x)f(y).$$

Then $f(x) = e^{-\mu x}$ for some $\mu > 0$.

Proof. f is continuous and additive, so by Cauchy's functional equation $f(x) = e^{-\mu x}$ for a constant μ ; strict decrease gives $\mu > 0$. $\square$

Meaning. The exponential is not an arbitrary choice. Binding adds up contributions from everything in a cell (Definition 4). If two things each hold q , then to leave, q must escape both, so the chance of escaping both should be the product of the chances of escaping each. The only well-behaved function that turns added binding into multiplied escape is an exponential. The constant μ sets how strong a hold must be to matter.

Example. Two independent holds, each of binding 1, with $\mu = 1$: each alone permits $e^{-1} \approx 0.37$ and together $e^{-2} = e^{-1} \cdot e^{-1} \approx 0.14$. The alternative $f(x) = 1/(1+x)$ would give $1/3$ for the two together, but $1/2 \cdot 1/2 = 1/4$ for the product of the two separately, so it fails to combine consistently.

Proposition III.6 (Stationary states of flow in a frozen environment). Fix q with $A^+(q, q) = 0$, and suppose all amounts other than q are held fixed, so that q is a fixed function on $\mathcal{X}$. Write $w(x) = e^{-\mu q(x)}$ and $u = w_q$. Then:

1. q is a fixed point of flow if and only if u is constant on $\mathcal{X}$.
2. If $\mu < 1$, iterating flow from any initial $q \geq 0$ converges to the unique fixed point with the same total, namely

$$q^*(x) = \sum_y q(y) \sum_y w(y)^{-1} w(x)^{-1}.$$

3. If $\mu > 2$, the convergence has no oscillating component.

Proof. Let Δ be the graph Laplacian of the grid, $(\Delta u)(x) = \sum_{y \sim x} u(x) - u(y)$, and $\mathcal{L} = \Delta w$. Flow is the linear map

$$q \mapsto \mathcal{L}^{-1} \Delta q.$$

(1) A fixed point satisfies $\mathcal{L}^{-1} \Delta q = q$. Since the grid is connected, Δq is the constants, so $u = q$ is constant.

(2) $\mathcal{L}$ is similar to the symmetric positive semidefinite matrix $\frac{1}{2} \Delta \frac{1}{2}$, so it is diagonalizable with real non-negative eigenvalues. Since $0 < w \leq 1$ and the spectrum of Δ lies in $[0, 4]$, every eigenvalue of $\mathcal{L}$ lies in $[0, 2]$. Hence the eigenvalues of $\mathcal{L}^{-1}$ lie in $[1/2, \infty]$, which for $\mu < 1$ is contained in $(-1, 1]$. The eigenvalue 1 has eigenspace $\mathbb{R} \cdot \mathbf{1} = \mathbb{R} \cdot \frac{1}{w}$, and all other components decay. Conservation (III.3) fixes the total, giving q^* .

(3) If $\mu > 2$, all eigenvalues lie in $[0, 1]$, so no component changes sign between iterations. $\square$

Meaning. This describes where free material ends up when only flow acts and nothing else changes. Flow settles when every pair of neighbors sends each other equal portions, which happens when amount times freedom to leave, $q e^{-\mu q}$, is the same everywhere. So material ends up concentrated where it is held most

strongly, in proportion to ϵ a strongly holding cell is a **trap**, like a sponge, rather than a barrier that turns material away. Part (2) guarantees that flow does settle, and part (3) shows why μ_{12} avoids material sloshing back and forth between neighbors. The assumption $A^+(q, q) = 0$ ensures that q 's binding does not depend on q 's own amount; the general case is open (Part XI).

Example. Three cells in a row, binding 2 in the middle and 0 at the ends, $\mu = 1$, and a total of μ . The middle cell releases q $\frac{1}{2}$ times more slowly than the ends, so in the limit it holds $\frac{1}{2}$ times as much as each end: approximately $(0.580, 0.840, 0.580)$. (Computed in Example VII.12.)

Part IV. Reactions

IV.1 Reacting pairs

Definition IV.1 (Reacting pairs in cells). For a state ν , let $P_{\mathbf{A}}(\nu)$ be the set of ordered pairs $\nu = (\mathfrak{q}, \mathfrak{r})$, with $\mathfrak{q}$ of type q and $\mathfrak{r}$ of type r , such that $\ell(\mathfrak{q}) > 0$, $\ell(\mathfrak{r}) > 0$, $A^+(q, r) + A^-(q, r) > 0$, and ν has one of the forms

Form	$\mathfrak{q}$	$\mathfrak{r}$	Condition
R1	$(\textcolor{teal}{\ast}q, x)$	$(\textcolor{teal}{\ast}r, x)$	$q \neq r$
R2	$(\textcolor{teal}{\ast}q, x)$	$(\textcolor{teal}{\ast}r, x)$	$(\textcolor{teal}{\ast}x) \neq \perp$
R3	$(\textcolor{teal}{\ast}q, x)$	$(\textcolor{teal}{\ast}r, x)$	$(\textcolor{teal}{\ast}x) \neq \perp$

Meaning. Within a cell, two amounts can react when the first one's key fits or clashes with the second one's lock. The pair is *ordered*: the first amount supplies the key, the second the lock, and each pair is considered in both orders separately, since a fit in one direction says nothing about the other. Three combinations are allowed: two different free types (R1), free with bundled (R2), and bundled with free (R3).

Example. A cell owned by a bundle, containing free q , free ν , the ambient, and bundled r , has the following candidate pairs; each is a reacting pair exactly when its fit or clash is positive.

Form	Candidate pairs (first, second)
R1	$(q, \textcolor{teal}{\ast}), (\textcolor{teal}{\ast}q), (q, \textcolor{teal}{\ast}), (\textcolor{teal}{\ast}q), (\textcolor{teal}{\ast}), (\textcolor{teal}{\ast})$
R2	(free q , bundled r), (free ν , bundled r), (r , bundled r)
R3	(bundled r , free q), (bundled r , free $\textcolor{teal}{\ast}$), (bundled r , $\textcolor{teal}{\ast}$)

Definition IV.2 (Contact pairs). For a set $\mathcal{A}$ of facing pairs (Definition 10), let $P_{\mathbf{A}}(\nu)$ be the set of ordered pairs with positive amounts and $A^+(q, r) + A^-(q, r) > 0$ of the forms

Form	$\mathfrak{q}$	$\mathfrak{r}$	Condition
R4	$(\textcolor{teal}{\ast}q, x)$	$(\textcolor{teal}{\ast}r, y)$	$(x, y) \in \mathcal{A}$

Form	$\mathfrak{s}$	$\mathfrak{q}$	Condition
R5	$(\mathfrak{s}q, y)$	$(\mathfrak{s}r, x)$	$(x, y) \preceq$

Meaning. When a bundle pushes against another, the material on either side of each facing pair touches, and can react, in both directions: the pusher's material acting on the other's (R4), and the other's acting on the pusher's (R5). This is how one bundle can wear another down by pressing against it.

Remark IV.3. Two bundled amounts in the same cell are owned by the same bundle, and no form pairs them; neither does any form pair a free amount with itself (R1 requires $q \neq r$). Fit between amounts of the same bundle therefore never produces a reaction. It contributes only to cohesion, links, and bond strength (Definition 6). Fit of a free type with itself contributes only to its binding (Definition 4).

Meaning. The same fit plays two different roles depending on where the amounts are. Inside one bundle, fit is **structure**: it holds the bundle together, and if it also caused reactions, every bundle would consume itself. Between separate things, fit is **interaction**: it causes reactions and holds free material in place.

Example. If w 's key fits $\mathfrak{s}$'s lock: when both are in the same bundle, they hold each other together and nothing reacts; when $\mathfrak{s}$ is bundled and w is free in the same cell, $\mathfrak{s}$ absorbs w .

IV.2 The reaction operator

For a pair $\mathfrak{r} = (\mathfrak{q}, \mathfrak{q})$ with types q and r , define:

Kind. $\mathfrak{r}$ is an *absorption* if $A^+(q, r) > 0$, and an *erosion* if $A^-(q, r) > 0$. Exactly one holds.

Raw quantity.

$$r^0(\mathfrak{r}) = \mathfrak{r}(\mathfrak{q}) \wedge \mathfrak{r}(\mathfrak{q}) \cdot A^+(q, r) + A^-(q, r) \cdot \mathfrak{r}(\mathfrak{q})$$

Consumption set. The amounts the reaction reduces, excluding ambient amounts:

$$\mathfrak{C}(\mathfrak{r}) = \mathfrak{q} \wedge \mathfrak{q} \cdot \mathfrak{r}(\mathfrak{q}, \mathfrak{q}) \cdot \mathfrak{r}(\mathfrak{q})$$

Destination. If $\mathfrak{q} = (\mathfrak{s}r, y)$ and $\mathfrak{r} \neq \perp$, then $\mathfrak{D}(\mathfrak{r}) = (\mathfrak{s}r, y)$; otherwise $\mathfrak{r}$ has no destination.

Meaning.

- **Kind.** A fit means **absorption**: the second amount takes in some of the first and converts it into its own product, without being used up itself, like a machine converting what is fed into it. A clash means **erosion**: the first amount wears down the second, turning part of it into the second's product, and is used up doing so. In both cases the direction is fixed by the order of the pair: the first amount offers its key and is taken in or spent; the second amount's lock is what absorbs, or what is eroded. In short, the key is offered, and the lock takes it in (see Table X.1 for examples).
- **Raw quantity.** How much would react, if there were no shortage. It is proportional to both amounts, since doubling either doubles the number of possible encounters between them (the law of mass action of chemical kinetics), and to the strength of the fit or clash. $\mathfrak{r}$ sets the overall speed.
- **Consumption set.** Which amounts the reaction reduces: always the first, and in erosion also the second. The ambient is never reduced (Axiom A3).

- **Destination.** Where the product goes: always next to the second amount, of the same form (free or bundled) and in the same cell. So when a bundled amount absorbs something, the product joins the bundle. If the second type's product is none, nothing is produced.

Example. Let x be owned by a bundle, with free q of amount 1, bundled r of amount 2, $A^+(q, r) = 0.5$, $r = r$ and $\perp = 0.1$. The pair (free q , bundled r) is an absorption with raw quantity $0.1 \cdot 2 \cdot 0.5 = 0.1$, consumption set free q , and destination bundled in x . (Computed in Example VII.13.)

Definition IV.4 (Reaction operator). For a state $\mathcal{A}$ and a set P of pairs, define:

1. **Demand:** $\bar{d}_x = \sum_{P \in \mathcal{P}} \bar{r}_x^0$.
2. **Factor:** $\ell_x = 1$ if $\bar{d}_x = 0$, else $\ell_x = \min(1, \bar{d}_x)$.
3. **Clamped quantity:** $\ell_x^0 = \bar{r}_x^0 \cdot \ell_x$.
4. **Update:** for every $\mathcal{A} \rightarrow \mathcal{A}'$

$$\ell_x' = \ell_x - \sum_{P \in \mathcal{P}} \ell_x^0 + \sum_{P \in \mathcal{P}} \ell_x^0.$$

Write $\mathcal{A}'$ for the resulting state. Then $\ell_x' = \ell_x^0$.

Meaning. Several reactions may draw on the same amount at once. The operator first totals how much all reactions would take from each amount (its **demand**). If the demand exceeds what is there, every reaction drawing on that amount is scaled down by the same **factor**, so the amount is used up exactly. A reaction that draws on two amounts uses the smaller of their two factors, so neither is overdrawn. Finally, every amount loses what reactions took from it and gains what reactions produced into it, all at once.

Example. An amount of 0.1 is the first amount of two absorptions, each with raw quantity 0.08. Its demand is 0.16, its factor is $0.1/0.16 = 0.625$, each reaction proceeds at 0.05, and the amount ends at exactly 0. (Example VII.14.)

IV.3 Properties

Proposition IV.5 (Well-definedness). $\mathcal{A}'$ is a state.

Proof. Non-negativity: for each $\mathcal{A}$, every $\mathcal{A}$ consuming $\mathcal{A}$ has $\ell_x \leq \bar{r}_x^0$, so the total subtracted is at most $\bar{d}_x \cdot \ell_x$. S5(1): ambient amounts are never in a consumption set, and never a destination, since products are never $\perp$. S5(2): bundled ambient amounts are 0, so never in a pair, and never a destination. S5(3): a destination $(\mathcal{A}, y)$ arises only when $\mathcal{A} = (\mathcal{A}, y)$ has $\ell_{\mathcal{A}} > 0$, which requires $\ell_y = 1$, y is unchanged. $\square$

Meaning. Reactions always produce a valid state: no amount goes negative, the ambient is untouched, and bundled material appears only in cells that a bundle owns.

Proposition IV.6 (Ambient; Axiom A3). $\mathcal{A}$ leaves $\mathcal{A}$ and $\mathcal{A}$ unchanged. $\square$

Meaning. The ambient takes part in reactions but is never consumed or produced.

Theorem IV.7 (Sources; Axiom A7). Let $P \subseteq \mathcal{P}$ be the absorptions whose first amount is ambient. Then

$$\ell_{\mathcal{A}}' = \ell_{\mathcal{A}} - \sum_{P \in P} \ell_{\mathcal{A}}^0.$$

In particular, $\mathcal{A}$ cannot increase unless $P \neq \emptyset$.

Proof. Each pair $\langle a, b \rangle$ changes Δ by $|a| - |b|$, since destinations are never ambient and consumption sets contain only non-ambient amounts. If $\langle a, b \rangle \in P$, then $b \neq \emptyset$: either a is non-ambient, or a is ambient and b is an erosion, in which case a is non-ambient (R1 requires distinct types). So the change is at most 0. If $\langle a, b \rangle \in P$, then $b = \emptyset$ and the change is at most 1. $\square$

Meaning. Each reaction's effect on the total amount of material is: plus what it produces, minus what it consumes. An absorption between non-ambient types consumes one amount and produces one, a one-for-one conversion. An erosion consumes two and produces at most one. Only an absorption of the ambient consumes nothing (the ambient is never reduced) while producing something. So the background is the only possible source of new material.

Reaction	Net change in Δ
absorption, product exists	0
absorption, no product	-1
erosion	-1
erosion, no product	-2
absorption of the ambient, product exists	$+1$

Proposition IV.8 (Locality; Axiom A1). Every pair in P_b involves a single cell; every pair in P_a involves a facing pair, which consists of two neighboring cells. Each destination lies in the cell of the pair's second amount. $\square$

Meaning. Reactions only happen between material that is touching.

Proposition IV.9 (Response as an absorption rate). Let $\zeta = (\zeta, q, x)$ with $\zeta(x) = \zeta$. The sum of raw absorption quantities over all R2 pairs whose second amount is ζ is

$$\zeta(x) \sum_{r \in \mathcal{R}} v(r) A^+(r, q).$$

In particular, if y is owned by no bundle, then $\zeta(y) = \sum_{r \in \mathcal{R}} v(r) A^+(r, q)$, and $\zeta(x) \zeta(y)$ is the raw rate at which $\zeta(x)$ would absorb free amounts if it were located at y .

Proof. Sum ζ over the R2 pairs $((r, x), \zeta)$ with $A^+(r, q) > 0$; pairs with $A^+(r, q) = 0$ contribute zero. For the second claim, $\zeta(y) = 0$ when $\zeta(y) = \perp$, so $\zeta(y) = v(y)$. $\square$

Meaning. Response is not just an abstract score: it is exactly how fast a bundled type would take in free material in a given cell. A drive attracted to high response is therefore attracted to wherever it would absorb the most.

Example. A bundled drive $\zeta = 1$ considers an unowned cell containing free Δ of amount 2 with $A^+(\zeta, q) = 0.9$ and the ambient with $A^+(\zeta, q) = 0.5$. Its response there is $2 \cdot 0.9 + 1 \cdot 0.5 = 2.3$, so with $\zeta = 0.1$ it would absorb at a raw rate of 0.23 per tick.

Part V. Bundle structure

μ is the composition of four operators, applied in order. Each evaluates its quantities at its own input.

Meaning. Reactions change what bundles contain: absorption adds products into them, and erosion turns their material into products. Either can leave a bundle holding material that no longer fits it, or weaken it until it should break. This stage checks every bundle and applies the consequences, in four steps.

Definition V.1 (Release). For every x with $(x) = \perp$ and every q with $\mu_q(x) > 0$ and $\mu_q^*(x) = 0$, set

$$\mu_q(x) \leftarrow \mu_q(x) + \mu_q(x), \quad \mu_q(x) \leftarrow 0,$$

simultaneously for all such (q, x) .

Meaning. A bundle keeps only what it holds. Any bundled material that nothing nearby in the bundle holds (zero local bond, Definition 7) falls out and becomes free, in the same cell. Without this rule, a product absorbed into a bundle would be trapped there forever, even if it fits nothing in the bundle.

Example. Erosion turns part of a wall made of w into rubble u , placing the rubble inside the wall. If u fits nothing in the wall and nothing in the wall fits u , then $\mu_u^* = 0$, and the rubble is released as free material, which begins to spread in the next tick.

Definition V.2 (Cleanup). For every x with $\mu_q(x) = 0$ for all q , set $(x) \leftarrow \perp$.

Meaning. A cell in which a bundle no longer has any material stops belonging to that bundle.

Definition V.3 (Split). For each bundle μ , let $\mu_\perp$ be the graph on μ with an edge x, y whenever $y \downarrow (x)$ and $(x, y) \geq \mu$. Let $\mu_1, \dots, \mu_n$ be its connected components. If $n > 1$, relabel each μ_i with a fresh label in $\mathcal{I}$.

Meaning. A bundle moves as one unit, so it should remain one piece only while its cells are attached to one another. Two neighboring cells count as attached when their connection is at least μ . If the attachments that remain leave the bundle in several separate groups, each group becomes a bundle of its own.

Definition V.4 (Dissolve). For each bundle μ with $\mu < \mu_\perp$ set $\mu_q(x) \leftarrow \mu_q(x) + \mu_q(x)$ and $\mu_q(x) \leftarrow 0$ for all q and $x \in \mu$; then set $(\mu) \leftarrow \perp$ on μ .

Meaning. A bundle whose overall bond strength is below the threshold is too weak to exist: all its material becomes free, and its cells become unowned.

Then

$$\mu \leftarrow \mu \circ \mu_\perp$$

Meaning. The order matters. Release comes first, so that material that fits nothing does not count toward attachments; cleanup next, so that empty cells do not form pieces of their own; split next, so that each piece is judged separately; and dissolve last, on the pieces that remain.

Example. A bundle of three cells in a row, each holding w with $A^+(w, w) = 0.8$ with the middle cell eroded to 0.25 and $\mu = 0.5$. Nothing is released, since w fits itself. Both connections to the middle are $0.4 < 0.5$ so the bundle splits into three pieces. The two end pieces have bond strength 0.8 and remain; the middle piece has bond strength 0.05 and dissolves. (Computed in Example VII.16.)

Proposition V.5 (Conservation). $\mathcal{B}$ preserves $\mathcal{V}_q(x)$ for every q and x .

Proof. Each operator only transfers amounts between $\mathcal{B}_q(x)$ and $\mathcal{B}_q(x)$ at the same (q, x) , or changes labels. $\square$

Meaning. Changing bundle structure never moves, creates, or destroys material; it only changes whether material is free or bundled, and which bundle it belongs to.

Proposition V.6 (Postconditions). After $\mathcal{B}$, every bundle $\mathcal{B}$ satisfies: $\mathcal{B} \neq \emptyset$; $\mathcal{B}$ is connected in $\mathcal{B}$ and $\mathcal{B} \geq \mathcal{B}$.

Proof. Cleanup removes cells with no bundled amounts. After split, each bundle is a connected component of its $\mathcal{B}$; relabeling does not change amounts, so its connections and $\mathcal{B}$ restricted to it are unchanged. Dissolve removes exactly the bundles with $\mathcal{B} < \mathcal{B}$ and does not alter any other bundle. $\square$

Meaning. After this stage, every bundle is guaranteed to be a single, sufficiently attached, sufficiently strong piece.

Proposition V.7 (Axiom A8). Every change made by $\mathcal{B}$ to a bundle $\mathcal{B}$ is determined by $\mathcal{V}_{\mathcal{B}, \mathcal{B}, \mathcal{B}}$ and $\mathcal{V}$, which depend only on $\mathcal{B}$'s own bundled amounts and their fits. $\mathcal{B}$ creates no bundle except by partitioning an existing one. $\square$

Meaning. Whether a bundle breaks depends only on itself; and this stage can divide bundles but never assemble new ones.

Part VI. Motion

Definition VI.1 (Motion). Let $\mathcal{B}_0$ be the set of bundles that have at least one drive at the start of $\mathcal{B}$. Draw a uniformly random ordering $\mathcal{B}_1, \mathcal{B}_2, \dots$ of $\mathcal{B}_0$. For $i = 1, 2, \dots$, with $\mathcal{B}_i$ the current state:

1. If $\mathcal{B}_i$ no longer exists or has no drive, skip it. Write $\mathcal{B}_i = \mathcal{B}_i$.
2. Let $\mathcal{B}_i$ be the valid displacements. For each $\delta \in \mathcal{B}_i$ compute the *desirability*

$$d(\delta) = \sum_{x \in \mathcal{B}_i} \mathcal{V}_q(x) - \mathcal{V}_q(x + \delta).$$

3. Draw δ^* uniformly from $\mathcal{B}_i(\delta)$.
4. **Stay:** if $\delta^* = (0, 0)$, do nothing.
5. **Blocked:** if δ^* is blocked, replace $\mathcal{B}_i$ by $\mathcal{B}_{\mathcal{B}_i(\delta^*)}(\mathcal{B}_i)$.
6. **Translate:** otherwise, replace $(\mathcal{B}_i)$ by $(\mathcal{B}_i)$, where for every q ,

$$\mathcal{V}_q(y) = \mathcal{V}_q(y - \delta^*) - \mathcal{V}_q(y - \delta^* + \delta^*) - \mathcal{V}_q(y - \delta^* + \delta^*) \mathcal{V}_q(y) \quad \text{if} \quad \mathcal{V}_q(y) = \mathcal{V}_q(y - \delta^* + \delta^*) - \mathcal{V}_q(y - \delta^* + \delta^*) \mathcal{V}_q(y) \quad \text{if}$$

$\mathcal{B}_i$ is unchanged.

Meaning. Bundles with drives take turns, in random order. On its turn, a bundle considers each of its possible moves, and for each one imagines having made it and asks how much its drives like what they would then occupy. The **desirability** adds, over every drive in every cell of the bundle:

- $g(x)$: how strongly that drive is felt (how much of it is there);
- q : whether what it responds to attracts (+1) or repels (−1) it;
- $g(x + \delta)$: how much of what it responds to is in the cell that part of the bundle would move into.

The bundle takes the best move, choosing at random among equally good ones. If it is to stay, nothing happens. If the move is blocked by another bundle, the bundle stays where it is, but the material across each facing pair reacts. Otherwise the whole bundle shifts one cell: its material moves with it, and the free material it passes over stays behind.

Processing bundles one at a time settles conflicts: if two bundles want the same cell, the first to move takes it, and the second finds its move blocked. The random order ensures no bundle is systematically favored.

Example. A single-cell bundle at (2, 2) holds an attracting drive of amount 2 and a repelling drive of amount 1. If the cell to the east contains the most of what attracts it and none of what repels it, east scores highest, and the bundle moves east. If the east cell belongs to another bundle, it stays, and its material reacts with the other bundle's material across the boundary. (Computed in Example VII.17.)

Proposition VI.2 (Well-definedness). Each sub-step maps $\mathcal{B}$ to $\mathcal{B}$.

Proof. Blocked moves apply τ , which preserves (IV.5). For translation: δ is valid, so $\delta \in \mathcal{B}$; the move is unblocked, so every cell of δ was owned by $\mathcal{B}$ or by no bundle, and no other bundle's cells are overwritten. Bundled amounts are written only to cells now owned by $\mathcal{B}$, and cells leaving $\mathcal{B}$ receive $\tau = 0$ and $\tau = \perp$, so S5(3) holds. $\square$

Meaning. Moving never overlaps two bundles, never pushes a bundle off the grid, and never leaves bundled material in an unowned cell.

Proposition VI.3 (Conservation under translation). Translation preserves $\sum_x g(x)$ and $q(x)$ for every q and x .

Proof. $y \mapsto y - \delta$ is a bijection from δ onto $\mathcal{B}$, and τ is unchanged off δ . $\square$

Meaning. A moving bundle carries all its material with it, losing and duplicating none, and leaves free material exactly where it was.

Proposition VI.4 (Axiom A9). The choice of δ depends only on the amounts g and valences q of $\mathcal{B}$'s drives (types with $q = 0$ contribute 0 to τ) and on their responses g at the cells of δ , together with the random ordering and tie-breaking. A blocked move does not change $\mathcal{B}$. $\square$

Meaning. The motion law is one concrete law satisfying Axiom A9.

Remark VI.5 (Exclusion of the bundle's own amounts). For a bundle occupying several cells, δ overlaps $\mathcal{B}$. Using τ instead of τ would make τ include the response of $\mathcal{B}$'s drives to $\mathcal{B}$'s own amounts, biasing τ toward overlap with itself. τ removes exactly those terms.

Meaning. Without the exclusion, a bundle would be attracted to its own material and would tend to stay put, or move only in ways that keep it overlapping its old position.

Remark VI.6 (Effects of contact deferred). Contact reactions in a blocked move may weaken either bundle. Their structural consequences are resolved by τ in the next tick.

Meaning. If pushing erodes a bundle, any resulting release, split, or dissolution happens at the bundle-structure stage of the following tick.

Part VII. Operation of the model

Parts III–VI define each operator in isolation. This part describes how the model runs as a whole: how ticks follow one another, what holds at every tick, the exact order in which a tick reads and writes the state, and worked computations of every operator.

VII.1 Runs

Definition VII.1 (External input). An *external input* is a map ι_τ such that $\iota_\tau(\cdot)$ differs from $\cdot$ only in the values of q for $q \neq \tau$ and of $\cdot$ and $\cdot$. The identity map is an external input, representing no intervention.

Meaning. An external input is any intervention by the player between ticks: placing or removing loose material or bundles. It may not touch the ambient.

Definition VII.2 (Run). Given an initial state $\theta \in \mathcal{S}$ a sequence of external inputs $(\iota_\tau)_{\tau \geq 0}$, and independent randomness for each tick, the *run* is the sequence

$$\theta_{\tau+1} = \tau(\iota_\tau), \quad \tau = 0, 1, 2, \dots$$

A run *without input* is one with $\iota_\tau = \text{id}$ for all τ .

Meaning. A run is the whole history of a world: start from an initial state, and at each tick let the player intervene (or not), then apply one tick of the dynamics. Since motion is random, a run is a random sequence of states; with fixed seeds, it is determined.

Proposition VII.3 (Runs remain in the state space). $\theta_\tau \in \mathcal{S}$ for every τ .

Proof. τ maps into $\mathcal{S}$ by definition. τ preserves non-negativity (III.2) and changes only q for $q \neq \tau$; τ maps into $\mathcal{S}$ (IV.5). Release, cleanup, split, and dissolve preserve S5 by inspection: bundled amounts are only set to zero or relabeled, and $\cdot$ is set to $\perp$ only on cells where $\cdot$ is zero. τ maps into $\mathcal{S}$ (VI.2). Induction on τ . $\square$

Meaning. However long a world runs, and whatever the player does, it never reaches an invalid state.

Proposition VII.4 (Invariants along a run). For every τ :

1. $\cdot \geq 1$ and $\cdot \geq 0$;
2. all amounts are non-negative;
3. for every type q , the total $\sum_x \tau(x)$ changes during a tick only through τ and contact reactions in τ .

Proof. (1) and (2) are part of VII.3. For (3): flow preserves $\sum_x q(x)$ (III.3) and does not touch $\cdot$; bundle structure preserves each $\tau(x)$ (V.5); translation preserves each $\sum_x \tau(x)$ and each $q(x)$ (VI.3). The remaining changes are reactions. $\square$

Meaning. Three things hold at every tick of every run: the background is untouched, nothing is negative, and spreading, bundle changes, and movement never create or destroy any type. Only reactions do.

Corollary VII.5 (No growth without sources). Suppose no type r satisfies both $A^+(r) > 0$ and $r \neq \perp$. Then along any run without input, $(\tau+1) \models \neg \psi$.

Proof. By VII.4(3), only reactions can change ψ . In the proof of Theorem IV.7, every pair outside P changes ψ by at most 0, and every pair in P changes it by (τ, τ) . A pair in P has the form $((\tau, x), \alpha)$ with α of type r and $A^+(r) > 0$; by hypothesis $r = \perp$, so it has no destination. Contact pairs involve only bundled amounts, which are never ambient, so none is in P . $\square$

Meaning. A world without any source type can only lose material over time, never gain it.

Example. In a world whose only reactions are erosions and the ambient draining free types, the total amount of material decreases every tick in which any reaction occurs.

Proposition VII.6 (Translation preserves bundle structure). If bundle $\mathcal{A}$ translates by δ , then for all $x, y \in \mathcal{A}$ with $y \prec x$ and all $\mathcal{B}$ the translated bundle $\mathcal{A} + \delta$ satisfies

$$(\mathcal{A} + \delta)(x) = \mathcal{A}(x), \quad (\mathcal{A} + \delta)(x, y) = \mathcal{A}(x, y), \quad (\mathcal{A} + \delta)(x) = \mathcal{A}(x), \quad \mathcal{B} \models \psi$$

Proof. $(\mathcal{A} + \delta)(x) = \mathcal{A}(x)$ for $x \in \mathcal{A}$. Since $\|(y + \delta) - (x + \delta)\|_1 = \|y - x\|_1$, the map $x \mapsto x + \delta$ is a bijection from $\mathcal{A}$ onto $\mathcal{A} + \delta$ that preserves the neighbor relation. Each quantity is a sum over cells and neighbor pairs of the bundle, so each is carried over term by term. $\square$

Meaning. Moving a bundle does not weaken or strengthen it: since all of its material shifts together and touching cells still touch afterward, every measure of how well it holds together is the same after the move.

Corollary VII.7 (Bundle integrity at tick boundaries). In $\tau+1$, every bundle that took part in no contact reaction during the motion step of tick τ satisfies the conclusions of Proposition V.6: its cell set is non-empty and connected in its graph of connections at least τ , and its bond strength is at least τ .

Proof. The conclusions hold after τ (V.6). During τ a bundle that takes part in no contact reaction is either unchanged or translated, and translation preserves every quantity in V.6 (VII.6). $\square$

Meaning. At the end of every tick, every bundle is guaranteed to be whole and strong enough, unless it was in a collision during that tick; in that case, the next tick's bundle-structure stage settles its condition.

Bundles that did take part in a contact reaction may violate these conclusions until $\tau+1$ restores them (Remark VI.6).

VII.2 The tick as a procedure

The following procedure computes $\mathcal{S} := (\mathcal{S}^i)$. Each line states which intermediate state its quantities are evaluated at.

Procedure VII.8 (One tick). Input: $\mathcal{S}^{(0)} = \mathcal{S}$

1. **Flow.** From $\mathcal{S}^{(0)}$, compute the totals $\tau(x)$, the bindings $\alpha(x)$, and the shares $\eta(x)$ for every $q \in \mathcal{Q}$. Apply Definition III.1. Call the result $\mathcal{S}^{(1)}$.
2. **Reactions.** From $\mathcal{S}^{(1)}$, compute the set $P(\mathcal{S}^{(1)})$, and for each pair its kind, raw quantity, consumption set, and destination. Compute demands, factors, and clamped quantities. Apply the update of Definition IV.4. Call the result $\mathcal{S}^{(2)}$.

3. **Release.** From $\mathcal{G}^{(2)}$, compute $\varphi_q(x)$ for every owned cell and type. Release every positive bundled amount with zero local bond. Call the result $\mathcal{G}^{(3)}$.
4. **Cleanup.** In $\mathcal{G}^{(3)}$, set $\varphi(x) = \perp$ on every cell with no positive bundled amount. Call the result $\mathcal{G}^{(4)}$.
5. **Split.** From $\mathcal{G}^{(4)}$, compute every connection $\varphi(x, y)$, form each graph $\mathcal{G}_i$ and relabel components. Call the result $\mathcal{G}^{(5)}$.
6. **Dissolve.** From $\mathcal{G}^{(5)}$, compute every bond strength φ_{ij} and dissolve every bundle with $\varphi_{ij} < \varphi_{ij}^*$. Call the result $\mathcal{G}^{(6)}$.
7. **Motion.** From $\mathcal{G}^{(6)}$, determine the bundles with drives and draw their order. For each in turn, compute φ_{ij}^* and φ_{ij} from the *current* state, draw δ^+ , and stay, react across facing pairs, or translate. The final state is $\mathcal{G}^{(7)}$.

Steps 1 and 2 are simultaneous over cells: no quantity in them is computed from a partially updated state. Step 7 is sequential over bundles: each bundle's decision is computed from the state left by the bundles processed before it.

Meaning. This is the tick written as a recipe, naming each intermediate state, so there is no ambiguity about which values each calculation uses.

Table VII.9 (Reads and writes).

Step	Reads	Writes
Flow	φ, φ_{ij}	q for $q \neq \varphi$
Reactions	$\varphi, \varphi_{ij}, \varphi_{ij}^*$	φ, φ_{ij}
Release	φ, φ_{ij}	φ, φ_{ij}
Cleanup	φ	φ
Split	φ, φ_{ij}	φ (labels only)
Dissolve	φ, φ_{ij}	$\varphi, \varphi_{ij}, \varphi_{ij}^*$
Motion	$\varphi, \varphi_{ij}, \varphi_{ij}^*$	φ, φ_{ij} by translation; φ, φ_{ij} by contact reactions

Remark VII.10 (Propagation of effects). Table VII.9 determines how a change propagates. A product created by a reaction at tick t :

- is subject to release in the same tick, if it is bundled and has zero local bond;
- affects motion in the same tick, through φ_{ij} if it is free or belongs to another bundle, and through φ_{ij}^* if it is a drive of a bundle;
- affects flow, as a moving free amount or through binding, only from tick $t + 1$, since flow precedes reactions.

Likewise, a bundle that translates at tick t grips and reacts with the free amounts of its new cells from tick $t + 1$.

VII.3 Worked computations

In each example, all fits and clashes not stated are zero, and the ambient takes part in no reaction.

Example VII.11 (Flow). Let $m = 2$, $\varphi = 1$, so $\varphi = \varphi_1, \varphi_2$ with $\varphi(x_1) = \varphi_2$. Let q have $\varphi = 0$, and $\varphi = 1$. Starting from $q = (10, 0)$:

Tick	$q(x_1)$	$q(x_2)$	$q(x_1)$	$q(x_2)$
0			10	0
1	2.5	0	$10 - 25 + 0 = 5$	$0 - 0 + 25 = 25$
2	1.875	0.625	6.25	3.75

Each x_i retains the three shares directed outside i . The total is 10 at every tick (III.3), and by III.6 the run converges to $(\frac{1}{3}, \frac{1}{3})$.

Example VII.12 (Flow with binding). Let $m = 3$, $i = 1$, with x_2 the middle cell, $i = 1$, $r = 1$, and $q = (0, 0)$. Let x_2 be owned by a bundle holding $w(x_2) = \frac{1}{3}$ with $A^+(q, w) = 0$ and $A^+(q, q) = A^+(q, \frac{1}{3}) = 0$, so $q = (0, 2, 0)$. Considering μ alone:

Tick	$q(x_1)$	$q(x_2)$	$q(x_3)$
1	6	2	0
2	4.568	3.364	0.068
∞	0.852	6.296	0.852

The limit is $\frac{1}{3} w^{-1} = (1, \frac{2}{3}, 1)$ with total $\frac{1}{3}$ (III.6). Proposition III.6 guarantees convergence for $\mu < 1$; here $\mu = 1$ also converges, because the eigenvalues of μ for a path of three cells lie in $[0, 3]$, so those of $\frac{1}{3}\mu$ lie in $[0, 3]$.

Example VII.13 (Absorption and erosion). Let x be owned by a bundle, with $q(x) = 1$, $r(x) = 2$, $r = r$ and $r = 0.1$. Then $P_k = \mu$ with $\mu = ((r, q, x), (r, x, x))$, an R2 pair.

Absorption, $A^+(q, r) = 0.5 \cdot \frac{0}{r} = 0.1 \cdot 1 \cdot 2 \cdot 0.5 = 0.1$; $\mu = (r, q, x)$; $\mu(q, x) = 0.1$; $r = 1$; $\mu = 0.1$; $\mu = (r, x)$.

Erosion, $A^-(q, r) = 0.5$ identical, except $\mu = (r, q, x), (r, x, x)$.

	$q(x)$	$r(x)$	$r(x)$	Change in μ
Before	1	2	0	
After absorption	0.9	2	0.1	0
After erosion	0.9	1.9	0.1	-0.1

Both changes in μ agree with the proof of Theorem IV.7: $\mu(1 - |\mu|)$.

Example VII.14 (Clamping). Let $q(x) = 0.1$, and suppose P_k contains two absorptions μ_1, μ_2 with first amount (r, q, x) and $\frac{0}{r}(\mu_1) = \frac{0}{r}(\mu_2) = 0.08$. Then

$$\mu(q, x) = 0.16 \quad (r, q, x) = 0.1016 = 0.05 \quad (\mu_1) = (\mu_2) = 0.05$$

and $\mu(x) = 0.1 - 0.05 - 0.05 = 0$, as guaranteed by IV.5.

Example VII.15 (Source). Let $v(x) = 2$, $A^+(r) = 0.5$, $r = u$, and $\alpha = 0.1$. Then $\alpha = ((r, x), (r, x))$ is an R1 absorption with $\emptyset = \emptyset$, so $\ell_j = \ell_j^0 = 0.1 \cdot 1 \cdot 2 \cdot 0.5 = 0.1$, and $u(x)$ increases by 0.1 while $\ell(x)$ and $v(x)$ are unchanged. α increases by $0.1 = \sum_{j \in \mathcal{J}} \ell_j$, so the bound of Theorem IV.7 holds with equality.

Example VII.16 (Release, split, dissolve). Let $m = 3$, $\alpha = 1$, $\beta = 0.5$ and let a bundle own all three cells, holding one type w with $A^+(w, w) = 0.8$ and $w = (1, 0.25, 1)$.

Release: $\alpha_j(x) > 0$ for every j , since w fits itself, so nothing is released. (If the bundle also held a type u with $A^+(u, j) = A^+(j, u) = 0$ for all j , then $\alpha_u = 0$ and u would be released.)

Split: $(x_1, x_2) = 1 \cdot 0.25 \cdot 0.8 + 0.25 \cdot 1 \cdot 0.8 = 0.4 < \alpha$, and likewise $(x_2, x_3) = 0.4$, has no edges, so α splits into three single-cell bundles.

Dissolve:

Bundle	Bond strength	Result
$\mathcal{P}_1$	$1 \cdot 1 \cdot 0.8 = 0.8$	remains
$\mathcal{P}_2$	$0.25 \cdot 0.25 \cdot 0.8 = 0.05$	dissolves: $w(x_2) = 0.25$, $w(x_2) = 0$, $\ell(x_2) = \perp$
$\mathcal{P}_3$	0.8	remains

The resulting bundles satisfy V.6.

Example VII.17 (Motion and contact). Let bundle own the single cell $x = (2, 2)$ and hold drives q with $\eta = +1$, $\ell(x) = 2$, and $\alpha = -1$, $\ell(x) = 1$. Suppose the excluded responses at the five target cells are:

δ	$x + \delta$	α_j	α_j	$\ell(\delta) = 2 \cdot \alpha_j - \alpha_j$
$(0, 0)$	$(2, 2)$	0.1	0.1	0.1
$(1, 0)$	$(3, 2)$	0.5	0	1.0
$(0, 1)$	$(2, 3)$	0.3	0.6	0
$(-1, 0)$	$(1, 2)$	0	0	0
$(0, -1)$	$(2, 1)$	0	0	0

The argmax is unique, so $\delta^* = (1, 0)$.

Unblocked: if $\ell((3, 2)) = \perp$, then α translates: $\alpha((3, 2)) = 2$, $\ell((3, 2)) = 1$, $\alpha((3, 2)) = \alpha$ and $(2, 2)$ becomes unowned with no bundled amounts.

Blocked: if $\ell((3, 2)) = \alpha$ then $\ell(\delta) = \ell((2, 2), (3, 2))$. Suppose α also holds $\ell((2, 2)) = 1$, α holds $w((3, 2)) = 2$, $A^-(w) = 0.5$, $w = u$, and $\alpha = 0.1$. The R4 pair $((\alpha, (2, 2)), (\alpha, w, (3, 2)))$ is an erosion with $\alpha = 0.1$, giving

$$\ell((2, 2)) = 0.9 \quad w((3, 2)) = 1.9 \quad \alpha((3, 2)) = 0.1,$$

and α remains at $(2, 2)$. If $A^+(u, j) = A^+(j, u) = 0$ for every type j held by α then at tick $\alpha + 1$ release (V.1) frees the 0.1 of u at $(3, 2)$.

Part VIII. Invariance

This part proves that the model has no hidden special cases and no meaningful coordinates: only the relationships between types matter.

VIII.1 Relabeling of types

A *type automorphism* is a bijection α with $\alpha(\perp) = \perp$ such that, for all q, r ,

$$A(q, r) = A(\alpha(q), \alpha(r)), \quad \alpha(\alpha^{-1}(q)) = q, \quad \alpha^{-1}(\alpha(r)) = r,$$

with $\alpha(\perp) = \perp$. It acts on states by $\alpha(x) = (\alpha^{-1}(r), \alpha^{-1}(p))$, that is, $\alpha(x)(x) = q(x)$.

Meaning. A type automorphism is a way of renaming types that changes nothing observable: every renamed type fits, clashes, produces, and pulls exactly as the original did.

Theorem VIII.1 (Uniformity; Axiom A2). For every type automorphism α and state ν , $\alpha(\nu)$ and ν have the same distribution.

Proof. Every quantity in Parts III–VI is a finite sum over types of products of amounts with values of A^+ , ν , and ν or a condition on such quantities. Substituting α and reindexing each sum by $q \cdot \alpha$ shows $\alpha(\alpha(\nu)) = \nu$, $\alpha(\alpha(\nu)) = \nu$, and likewise for $\alpha(\nu)$ and α . Reacting pairs correspond under α with equal raw quantities, consumption sets, and destinations (the latter because $\alpha(\alpha^{-1}(r)) = r$), so demands, factors, and clamped quantities correspond. Bundle structure depends only on these quantities and on ν . Desirabilities correspond for each δ , so the argmax sets coincide, and the random choices can be coupled. Each operator therefore commutes with α , and so does their composition. $\square$

Meaning. If two worlds differ only by such a renaming, they evolve identically (up to the same renaming). This is the precise sense in which the model has no rules for particular types: every law sees only fits, clashes, products, and valences.

Example. Suppose types q and q' have identical fits and clashes with every type, in both directions, and the same product and valence. Swapping the names q and q' is a type automorphism, so a world with q on the left half of the grid and q' on the right evolves exactly like the world with q' on the left and q on the right, with the two names exchanged throughout.

VIII.2 Change of basis in signature space

Proposition VIII.2 (Basis invariance). Let $\alpha \in (d, \mathbb{R})$, and replace every key and lock by

$$k'_q = \alpha k_q, \quad l'_q = \alpha^{-1} l_q.$$

Then $A(q, r) = A(q, r)$ for all q, r , and hence $\nu = \alpha \nu$.

Proof. $k'_q \cdot l'_r = k_q \cdot \alpha^{-1} l_r = k_q \cdot l_r$, and α depends on keys and locks only through A . $\square$

Meaning. The coordinates of signature space are only a convenience. Any consistent change of coordinates (transforming every key by α and every lock by the inverse transpose of α) leaves every affinity, and therefore

everything the model does, exactly the same.

Example. With $d = 2$, let $k_q = (1, 0)$ and $l_r = (1, 0)$, so $A(q, r) = 1$. Take $\alpha = 2$, so $\alpha k_q = (2, 0)$ and $\alpha l_r = (2, 0)$, and $A(q, r) = 2 \cdot 2 = 1$. A rotation, for which $\alpha = 1$, likewise rotates all keys and locks together and preserves every affinity.

Consequence. Individual coordinates, lengths, and angles of keys and locks carry no meaning. Only the bilinear pairing between keys and locks is observable. In particular, the model is unchanged by any rotation of $\mathbb{R}^d$ applied simultaneously to all keys and locks.

VIII.3 Scaling

Proposition VIII.3 (Scale equivalence). Let $\alpha > 0$. The model with keys and locks $\alpha k_q, \alpha l_q$ and constants $(\frac{\alpha}{\psi}, \frac{\alpha}{\mu}, \frac{\alpha}{\nu})$ has the same dynamics as the original model with constants $(\frac{\alpha}{\psi}, \frac{\alpha}{\mu}, \frac{\alpha}{\nu})$.

Proof. Scaling multiplies every affinity by α ; with dead zone $\frac{\alpha}{\psi}$ every fit and clash is also multiplied by α . Then binding, response, cohesion, links, connections, bond strength, and local bonds are all multiplied by α . In flow, $\frac{\alpha}{\mu} = \frac{\alpha}{\mu}$. In reactions, $\frac{\alpha}{\nu} A^+ = (\frac{\alpha}{\nu}) A^+$. In bundle structure, $\frac{\alpha}{\nu} \geq \nu \iff \frac{\alpha}{\nu} \geq \nu$, likewise for μ and $\frac{\alpha}{\mu} = 0 \iff 0$. In motion, ν is multiplied by α , which leaves its argmax unchanged. $\square$

Meaning. Making all keys and locks longer is equivalent to adjusting the constants: binding, reaction speed, and the bond threshold absorb the change. So the overall size of the vectors is not an independent setting, and can be fixed once while tuning μ, ν , and ψ .

Example. With $\alpha = 2$, every affinity is multiplied by α . A world with doubled vectors and dead zone $\frac{1}{2}$ behaves exactly like the original world with μ and ν multiplied by $\frac{1}{2}$ and ψ divided by $\frac{1}{2}$.

The scale of the vectors and the constants μ, ν, ψ are therefore not independent parameters.

Part IX. Remarks on interpretation

IX.1 Self-affinity is not automatic. A type's key and lock are independent vectors (S4), so $A(q, q) = k_q \cdot l_q$ may be positive, zero, or negative. Belonging to the same type does not imply a fit. Self-affinity has two observable consequences:

- If $A^+(q, q) > 0$, free q contributes to its own binding, so by Proposition III.6 it tends to concentrate where it already is.
- If $A^+(q, q) > 0$, a bundle consisting of q alone has positive cohesion and can satisfy $\frac{\alpha}{\nu} \geq \nu$.

A type with $A^+(q, q) = 0$ neither clumps nor forms a bundle by itself.

IX.2 One fit, two roles. The same quantity $A^+(q, r)$ appears in two disjoint places: within a bundle it enters cohesion, links, and bond strength (Definition 6) and never produces a reaction (Remark IV.3); between amounts not in a common bundle it produces absorption (Part IV) and binding (Definition 4).

IX.3 Mass action. The raw quantity ϕ is bilinear in the two amounts, as in the law of mass action of chemical kinetics. The same bilinear form appears in cohesion and links, where it counts unit-to-unit pairings.

IX.4 Sources and sinks. By Theorem IV.7, a type r with $A^+(r,r) > 0$ and $r \perp$ is the only possible source of material, producing $A^+(r,r)$ of r per tick from an amount of r . A type q with $A^+(q, \cdot) > 0$ decays geometrically, losing the fraction $A^+(q, \cdot)$ per tick in the absence of other reactions.

Part X. Example

This part assembles all the mechanisms in one small world, to show how behaviors that no rule mentions arise from four tuples.

Take $d = 2$, $\omega = 01$, $\iota = 01$, $\rho = 1$, $\omega = 05$ and $\omega = \omega$.

Type	k	l	r	v
ω	(0, 1)	(03, 0)	$\perp$	0
ω	(0, 1)	(0, 0)	ω	0
ω	(1, 0)	(0, 0)	$\perp$	0
ω	(0, 1)	(1, 0)	$\perp$	+1

The relevant fits are

$$A^+(\omega, \omega) = A^+(\omega, \omega) = A^+(\omega, \omega) = 05 \quad A^+(\omega, \omega) = 02, \quad A^+(\omega, \omega) = 09$$

$$A^+(\omega, \omega) = A^+(\omega, \omega) = A^+(\omega, \omega) = 0.$$

Table X.1 (How the fits act). In each fit, the type whose **lock** fits is the one that absorbs, and the type whose **key** fits is absorbed (Part IV.2):

Fit	Who absorbs whom	Effect
$A^+(\omega, \omega) = 09$'s key fits ω 's lock	ω absorbs ω	ω removes ω since $\omega \perp$
$A^+(\omega, \omega) = 02$: ω 's key fits the ambient's lock	the ambient absorbs ω	ω decays, since $\omega \perp$
$A^+(\omega, \omega) = 05$ the ambient's key fits ω 's lock	ω absorbs the ambient	ω produces ω every tick, since the ambient is never reduced

Since ω 's lock is (0, 0), nothing fits it, and ω absorbs nothing. The self-fits $A^+(\omega, \omega)$ and $A^+(\omega, \omega)$ cause no absorption at all: within a bundle, fit holds material together rather than producing reactions (Remark IV.3).

Consider a row of cells with a single-cell bundle containing ω at one end and a single-cell bundle containing ω several cells away. Then:

- ω is a source of ω (IX.4): each tick, 005 of ω is produced inside the bundle containing ω
- $\omega \perp$ in that bundle, so the produced ω is released as free (V.1).

3. Free .has binding 02 everywhere (from the ambient alone), diffuses, and decays at rate 002 per tick, forming a decreasing profile away from the source.
4. .js a drive with $\text{.y} = +1$ and $A^+(\text{.y}) > 0$, so the bundle containing . moves up the profile of $\text{.}(VI.1)$, absorbing free .as it goes (IV.9).

None of these behaviors is specified by a rule; each is a consequence of the four tuples.

Part XI. Open problems

An *open problem*, in this document, is a question the axioms and definitions leave unresolved. There are two kinds:

- **Modeling gaps:** behaviors the model, as defined, provably cannot produce, but which a richer model might want. Closing one requires changing an axiom or adding a law.
- **Mathematical questions:** properties of the model as defined that have been neither proved nor disproved.

Neither kind affects the consistency of Parts I–VIII: every result stated there holds as proved.

XI.1 Modeling gaps

Proposition XI.1 (Owned cells never increase). Along any run without input, .own is non-increasing.

Proof. Flow and reactions do not change . . Release, split, and relabeling do not enlarge the set of owned cells; cleanup and dissolve can only shrink it. Translation maps . bijectively onto $\text{.} \neq \delta$, sets $\text{.} \neq \perp$ on the cells it leaves, and, being unblocked, overwrites no cell of another bundle; so the number of owned cells is unchanged. $\square$

Meaning. The region of the world occupied by bundles can only stay the same size or shrink. Bundles can move, split, and fall apart, but nothing ever makes the occupied region larger.

1. **Formation.** By XI.1 and Axiom A8, no free amount ever becomes bundled: bundled material can grow only by reactions whose destination is bundled (forms R2, R4, R5), never by capture of free material into structure. A formation law (condensation of free amounts into a new bundle, or merging of neighboring bundles) would require amending A8, and would need a formation threshold strictly above .to to avoid forming and breaking in alternate ticks.
2. **Repulsion.** The fraction sent from x to a neighbor depends only on $\text{.}_q(x)$ (Axiom A5). No property of the receiving cell can reduce what it receives, so a cell can retain material (III.6) but cannot reject it.
3. **Cost of motion.** Translation causes no reaction (Definition VI.1, step 6). A bundle interacts with the free amounts of its cells identically whether or not it moved, so movement itself has no physical consequence.
4. **Rigid, unit-speed bundles.** Bundles change shape only by losing cells or splitting, and translate by at most one cell per tick.
5. **Non-local decisions.** . depends on cells within distance 1 of . , which, for a large bundle, includes cells far from one another. This is consistent with Axiom A1, which constrains changes rather than decisions, but it is not local in the stronger sense of bounded range.

XI.2 Mathematical questions

- 6. **Boundedness of material.** If a type r has $A^+(r) > 0$ and $r = r$, then in a cell where r takes part in no other reaction, r is multiplied by $1 + A^+(r)$ every tick, so r is unbounded. Corollary VII.5 gives a sufficient condition for r to be non-increasing (no sources at all). *Open:* characterize, in terms of the type table, when r remains bounded along every run without input in the presence of sources.
- 7. **Long-run behavior.** Proposition III.6 describes the limit of flow alone in a frozen environment without self-binding. *Open:* existence, uniqueness, and stability of fixed points and periodic orbits of f (for a fixed realization of randomness), and the behavior of flow when $A^+(q, q) > 0$, where binding depends on q itself and the dynamics become nonlinear. In that case concentration can reinforce itself, and a condition for clumping (phase separation) in terms of r, r and $A^+(q, q)$ is not established.
- 8. **Sensitivity to constants.** *Open:* how the qualitative behavior of runs (persistence of bundles, formation of gradients, motion toward or away from sources) depends on r, r, r and r beyond the equivalences of Proposition VIII.3.

Appendix. Correspondence with the working specification

This document	Working specification
S1, S2	D1, D2
S3, S4	D3, D5–D13
S5	D4, D14, Axiom 1
S6	D0
Definitions 1–10	D8, D9, D14–D20 and the local bond of Step 4a
Definitions IV.1, IV.2, IV.4	D21
A1–A10	Axioms 2–11
Parts III–VI	Steps 2–5
Part VII	the tick overview, worked examples, and propagation of effects (Sections 4.4–4.6)

Changes made for rigor:

- 1. **Single-grid representation.** Bundles are encoded by the ownership function o and one bundled-amount function g ; this is equivalent to D14 because bundles are disjoint.
- 2. **Products are never ambient** (S4). Without this, a reaction could add to g , contradicting A3.
- 3. **The ambient is never bundled** (S5(2)).
- 4. **Randomness is explicit** in A9 and in μ , making μ a Markov kernel.
- 5. **Facing pairs are defined against all blocking bundles at once** (Definition 10), and structural effects of contact are deferred to the next tick (Remark VI.6).

6. **The order of motion** is drawn over bundles with drives at the start of the motion step, with bundles skipped if they cease to exist or lose their drives.
7. **Cleanup** of cells with no bundled amounts is performed before splitting, so that empty cells do not form components.